MAX30102

Heart-rate and blood-oxygen (pulse oximeter) sensor

The MAX30102 is a tiny optical sensor (5.6 x 3.3 mm) that shines red and infrared light into a fingertip or earlobe and measures how much comes back. From that light it works out the **pulse** and, with calibration, the **blood-oxygen level (SpO_2)**. It sends raw readings over I2C. Your code turns them into beats per minute.

Maker	Analog Devices (originally Maxim Integrated)
Measures	Reflected red and infrared light, for heart rate and SpO_2. Also has an on-chip temperature sensor (die temperature, +/-1 °C).
Light sources	Red LED about 660 nm, infrared LED about 880 nm. LED current 0 to 50 mA, set in software.
Converter	Up to 18-bit ADC. Sample rate 50 to 3200 samples per second. Pulse width 69 to 411 microseconds.
Buffer	FIFO that holds 32 samples, so the microcontroller can read in bursts.
Interface	I2C, address 0x57. Our wiring: SDA to GPIO21, SCL to GPIO22.
Power	Chip core 1.8 V, LEDs 3.3 V (3.1 to 5 V). Breakout boards add regulators. We power ours from 3.3 V.

How it works

- **1. Light in.** The LEDs shine into the finger.
- **2. Blood absorbs.** Blood absorbs part of the light. Skin, bone and tissue absorb a steady amount.
- **3. Light back.** The photodiode measures what is reflected. With every heartbeat the blood volume in the vessels changes, so the reading rises and falls slightly. This wave is called a **PPG** (photoplethysmogram).
- **4. Count.** The time between two peaks is one beat, so **BPM = 60 / T**.
- **5. SpO_2.** Oxygen-rich and oxygen-poor blood absorb red and infrared light differently. The ratio $R = (AC_{red}/DC_{red}) / (AC_{ir}/DC_{ir})$ is mapped to SpO_2 with a calibration curve.

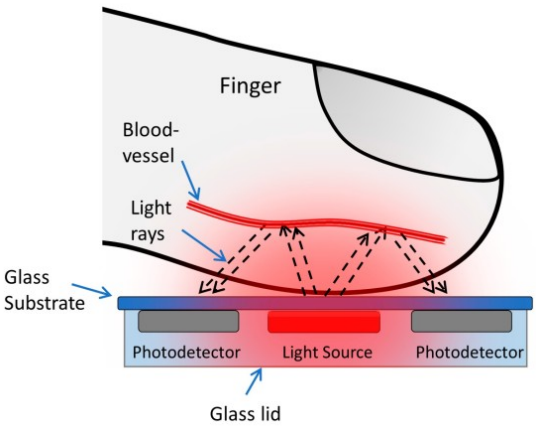


Fig 1. Light path through the finger: LED, blood vessel, photodetector.

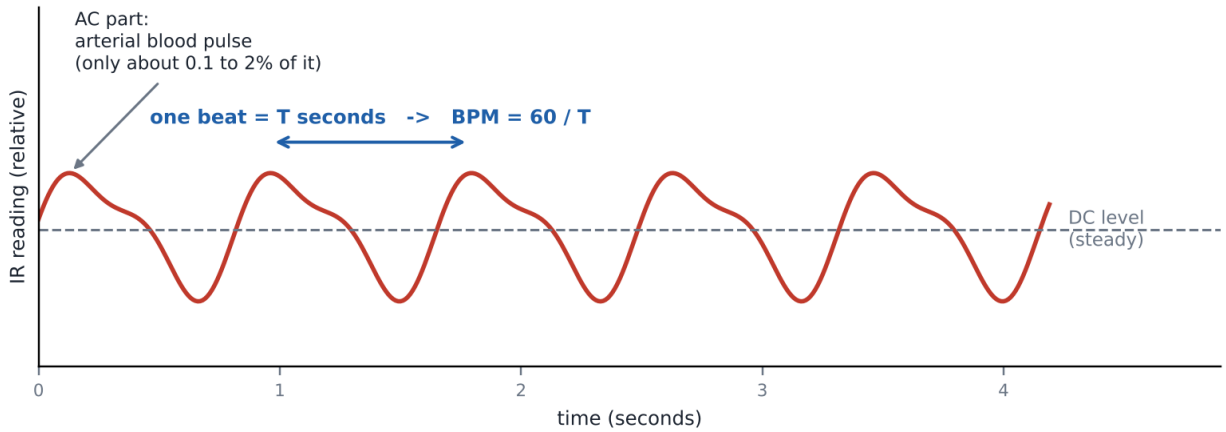


Fig 2. What the sensor sees: a steady DC level with a small pulse (AC) on top. Beat spacing gives the heart rate.

Inside the module, in one line

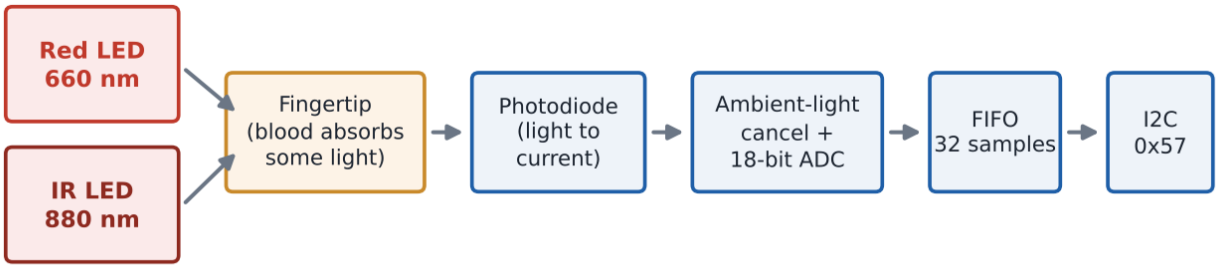


Fig 3. Signal path from LEDs to your microcontroller.

Pins and wiring

Pin	What it does	Connect to (ESP32 example)
VIN	Power input. Use 3.3 V (check your board before using 5 V).	3V3
GND	Ground	GND
SDA	I2C data line	GPIO21
SCL	I2C clock line	GPIO22
INT	Optional interrupt output (data ready, FIFO almost full). Not needed when polling.	Leave unconnected
IRD, RD	LED driver pins for the infrared and red LEDs. Not used in normal operation.	Leave unconnected

Only four wires are needed: VIN, GND, SDA and SCL. Breakout boards normally include the I2C pull-up resistors. The sensor shares the bus with other I2C devices as long as the addresses differ (MAX30102 is 0x57).

Where it is used today

Used today	Could be used for
• Smartwatches and fitness bands (pulse, SpO_2) • Smartphones and tablets (the datasheet lists these) • Fitness and wellness gadgets • Finger-clip pulse-oximeter style devices (hobby and consumer grade) • Student and maker projects	• Driver or operator fatigue monitoring (pulse trend, heart-rate variability) • Worker safety wearables in mines and plants (exertion, heat stress) • Health-screening kiosks and telehealth demos • Sports training zones and recovery tracking • Stress-aware apps, biofeedback and games • Elderly-care alerts for unusual heart rate

Know the limits

- **Motion ruins the signal.** A moving finger or loose contact causes false beats. Keep the hand still.
- **Contact matters.** Too little or too much pressure, cold fingers and strong ambient light all change the reading.
- **Not a medical device.** SpO_2 needs careful calibration. Use this for prototypes and trends, not diagnosis.
- **Strong signal needs the right software.** Some beat-detector libraries ignore very strong signals. We use our own detector for that reason.

Sources: MAX30102 datasheet (Analog Devices). Figures 2 and 3 are original diagrams.